



Lab 8: Functions

CSE 4108

Structured Programming I Lab

Lab Tasks

1. toLower() and beyond!:

We all know about the `tolower()` function from the `<ctype.h>` library, which converts an uppercase letter to a lowercase letter.

i. Write a function `void read_string(char ch[], int length)` that will read the input string.

ii. Write a function `char to_lowercase(char ch)` that will return a character in lowercase letter.

iii. Write a function `void print_string(char ch[], int length)` that will print the whole string in lowercase by repeatedly calling the `to_lowercase()` function on each character.

All these three functions should be used in your main function to take an input string and print the corresponding string in lowercase.

***DON'T** use the built-in `tolower()` function and any string related functions or format specifiers.

2. Poly-nom-nomial!:

Write a function called `polynomial(int x)` that computes the value of the following polynomial:

$$ax^7 + bx^6 - cx^5 - dx^4 + ex^3 - fx^2 + gx + 10$$

Here, a, b, c, d, e, f, g denotes the individual digits of your student id except the zeroes in the middle. For example, if your student id is 240042111, then the polynomial is:

$$2x^7 + 4x^6 - 4x^5 - 2x^4 + x^3 - x^2 + x + 10$$

Write a main function that asks the user to enter a value for x, calls the `polynomial()` function to compute the value of the polynomial, and then displays the value returned by the function.

3. Decimal-to-Binary and Binary-to-Decimal:

Write a function named `dec_to_bin(int d)` that takes a decimal integer as parameter and returns the corresponding binary value. Write another function named `bin_to_dec(int b)` that takes a binary integer as parameter and returns the corresponding decimal value.

In your main function, take a decimal number, and a binary number as input and print the converted values by calling the `dec_to_bin()` and `bin_to_dec()` functions.

Sample Input:

Input any decimal number: 65
Input any binary number: 1000001

Sample Output:

The Binary value is: 1000001
The Decimal value is: 65

4. Calculator:

Write a `calculator(int a, int b, char op)` function which takes two integers and a character representing an operation and returns the value after performing addition, subtraction, multiplication, division, remainder or exponentiation. Your main function should take two integers and a character as input and print the result by calling the `calculator()` function.

Sample input:

Enter two numbers: 10 5

Enter a choice: *

Sample output:

Result of the operation: 50

5. Total Marks:

You are given an array of floats, **marks**, denoting the marks scored by students in a class.

- The alternating elements marks[0],marks[2],marks[4] and so on denote the marks of boys.
- Similarly, marks[1],marks[3],marks[5] and so on denote the marks of girls.

Write a function "sum_of_marks":

Parameters:

- 1.**marks** array
- 2.Length of the array
- 3.Character variable ('B or b' for boys, 'G or g' for girls)

Returns: The summation of marks for all the boys or girls, based on the parameters.